

Abstract This study analyzed maternal mortality trends globally and regionally from 2000 to 2023. The goal of this analysis was to observe trends in Maternal Mortality Ratio (MMR), maternal deaths and lifetime risk (LTR) across the various regions. Linear regression coupled with percentage changes and variability were conducted to assess the disparity trends. This study highlights the global decline in MMR and maternal deaths, associated with the increase in LTR while also highlighting the regional disparities seen in all these factors.

Introduction Maternal mortality, which is the death of women during pregnancy, childbirth or shortly after has been a tragic but common occurrence throughout history. History has shown, specifically in the 19th century, that maternal mortality ratios (MMRs) in some countries were seen to be as high as 1000 deaths per 100000 live births. This represents the risk of 1 in 100 to 1 in 200 women dying due to any childbirth complication. Thankfully, this global statistic has decreased and it has been shown that by the mid-20th century, some high income countries have successfully reduced the deaths to under 10 per 100000 births. (Roser & Ritchie, n.d.)

This grand success is due to advances in medical care such as the advancement of antibiotics to treat illness and infections, as well as the improvement of maternal care before, during and after birth. Unfortunately, maternal mortality rates remain high in some countries, specifically mid to low-income countries where it has been shown that 95% of maternal deaths occur here. This specifically can be seen in sub-Saharan Africa region and some parts of Asia that still observe maternal health challenges due to only 73% of births in these regions are observed by healthcare professionals. (*Maternal Mortality*, 2025)

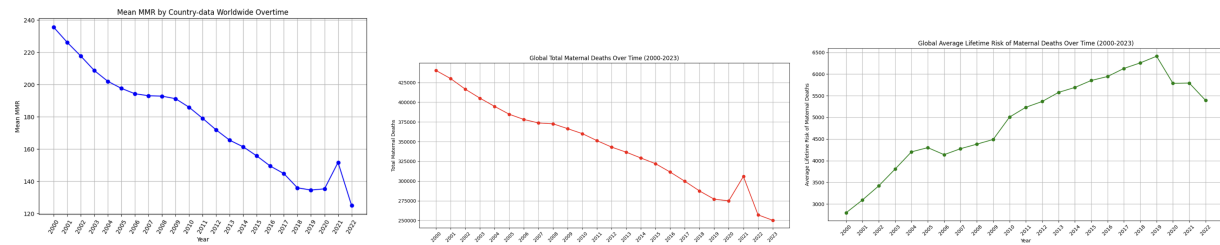
As global changes have occurred, maternal mortality is shown to still be a persistent issue, specifically in selected regions that are still observing these maternal health challenges. Specifically in 2023, an upward 700 women died daily from a child-birth issue, which equates to a death every two minutes worldwide. This issue requires our attention and additional passion towards improving skilled healthcare professions while also addressing the global healthcare barriers to attack inequalities that are disproportionately put onto women, specifically women in lower-income countries. (Roser & Ritchie, n.d.)

This study aims to analyze global and regional trends in maternal mortality from 2000 to 2023 as it assesses the disparity in MMR, maternal death and lifetime risk (LTR) across the different regions and provides insights into the factors that are a driving force in maternal health. Focusing on these trends allows the study to highlight the ongoing efforts performed to rescue maternal mortality and improve health care globally as a stepping stone into understanding the right direction towards improvement.

Methods Data for this analysis was sourced from UNICEF Maternal Mortality database (2000-2023). (*Trends in Maternal Mortality 2000 to 2023 - UNICEF DATA*, 2025) Global and regional disparities in MMR, maternal deaths and LTR were analyzed in Google Co-Lab. Linear regression models were applied to assess trends in these compounding factors, while percentage change and variability calculations were used across regional progresses to assess regional improvement or disparity. Data was categorized by UNICEF's regional classifications to explore disparities globally. Correlation analysis was assessed to

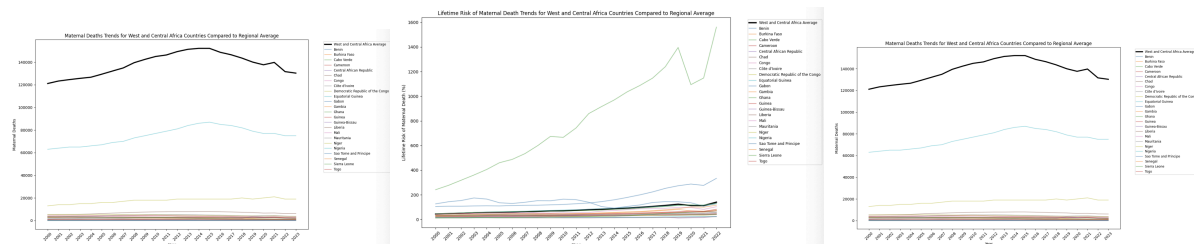
examine the relationship between MMR, maternal deaths and LTR. Visualization of all analysis and regression were provided to highlight trends in all data findings.

Results Global Results: MMR, Maternal Death, LTR

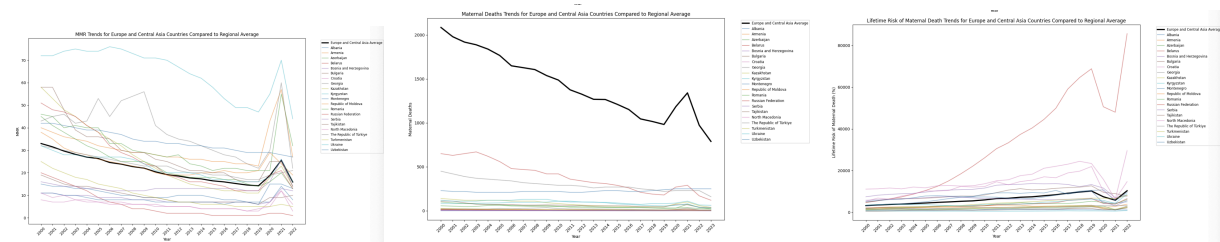


These global trends over time are illustrated to show the compounding factors on maternal health from 2000 to 2023. We can see that MMR and maternal death has shown a significant decrease over time while LTR showed an increase, however resulting from a slight decline at the end.

Trends In Africa Regions Countries: MMR, Maternal Death, LTR



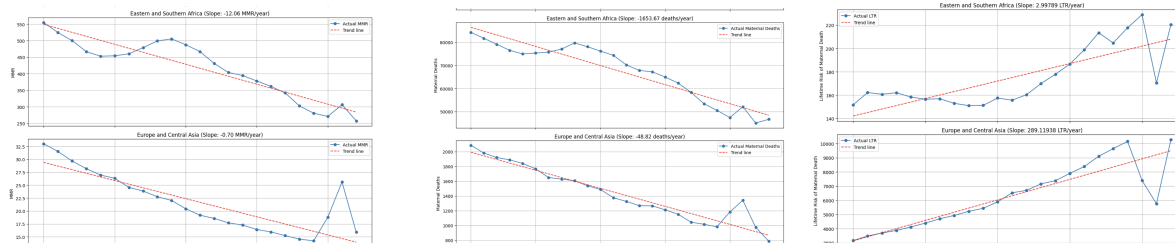
Trends In Europe and Central Asia Countries: MMR, Maternal Death, LTR



By comparing regional averages of maternal health indicators with trends from individual countries, we can easily visualize how certain countries are doing in relation to their region. Looking at Africa and Europe/Central Asia, you can really see the differences in maternal health. For example, in the African countries we looked at, MMR is much higher and hasn't shown much decline, whereas in Europe and Central Asia, MMR has been steadily decreasing. When we look at maternal deaths, countries in Europe and Central Asia show a decline, but African countries already had relatively low rates to start with, so there's not much change from 2000 to 2023. Finally, for Lifetime Risk (LTR), both regions have stayed pretty consistent, but African countries still have a higher regional average, meaning the risk for maternal death is higher in that part of the world.

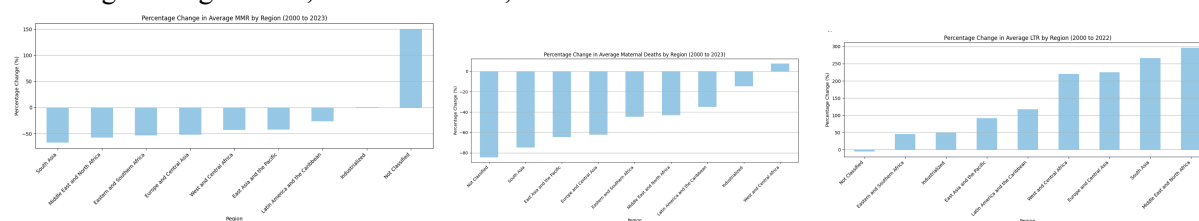
Linear Regression: MMR, Maternal Death, LTR

Eastern and Southern Africa - Top, Europe and Central Asia Countries - Bottom



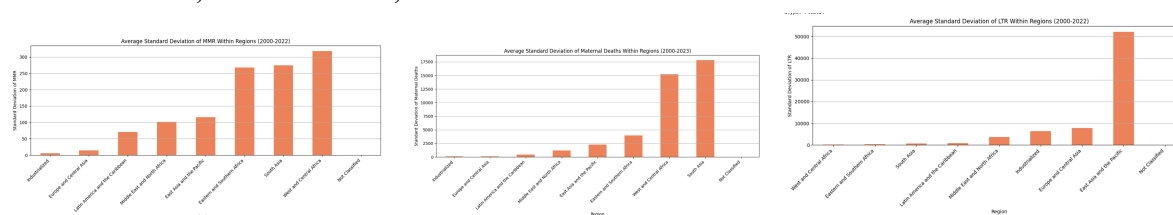
These plots of linear regression on the regional averages of MMR, Maternal Death, LTR reveal the annual rate of change in these trends to highlight improvements or disparities. We can see that regionally we cannot see much difference between these factors. MMR and maternal deaths show a steady decrease while LTR showed an increasing trend over time.

Percentage Change:MMR, Maternal Death, LTR



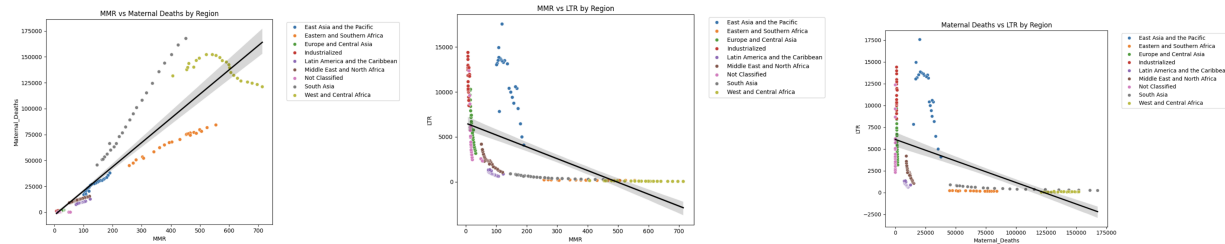
Percentage Change highlights the shift in maternal health indicators from 2000 to 2023 across regions, generally showing improvements in key maternal health factors. We observe a decline in MMR over time in most regions, with the exception of the non-classified cohort. Similarly, maternal deaths have decreased in most regions, though the West and Central Africa region stands out with an increase in maternal deaths over the same period. LTR has shown an upward trend across all regions, indicating a growing risk of maternal death in each area.

Variation: MMR, Maternal Death, LTR



Variation over time helps us understand the spread and diversity of data across regions. In the case of MMR, we observe high variation across most regions, suggesting that the data points are widely spread. This likely reflects significant disparities within certain countries in each region, pointing to uneven maternal health outcomes. For maternal deaths, we notice high variation in the regions of West and Central Africa and South Asia, indicating differences in mortality rates within these areas. LTR shows high variation only in the East Asia and Pacific region, highlighting significant differences in risk across countries in these regions.

Correlation: MMR vs Maternal Death, MMR vs LTR, Maternal Deaths vs LTR



These correlation plots illustrate the relationships between MMR, maternal deaths, and LTR, providing a visual representation of how these factors influence each other. For the relationship between MMR and maternal deaths, we observe a strong positive correlation, marked by an upward slope and tightly clustered data points. This trend is noticeable in regions such as East Asia and the Pacific and Middle East and North Africa, where increases in MMR are closely tied to higher maternal deaths. In contrast, the relationship between MMR and LTR shows a negative correlation with a looser data cluster representing a weaker relationship across most regions. Similarly, the correlation between maternal deaths and LTR also reveals a negative and weak correlation in most regions, suggesting that as maternal deaths decrease, the lifetime risk of maternal death doesn't significantly change or change consistently.

Multiple Regression

The multiple regression analysis for Europe and Central Asia reveals a strong relationship between maternal deaths and MMR. This relationship represents an increase in maternal deaths leading to a rise in MMR. The model explains 92% of the variation in MMR, indicating a strong fit. However, the analysis found no statistically significant effect of LTR on MMR. Despite the overall significance of the model (F-statistic = 114.3, p-value < 0.01), issues with residual autocorrelation (Durbin-Watson = 0.527) suggest further analysis is needed to refine the model. The multiple regression analysis for Eastern and Southern Africa shows a strong relationship between maternal deaths and MMR, with an increase in maternal deaths leading to a rise in MMR. The model explains 98.5% of the variation in MMR, indicating an excellent fit. However, the analysis found no statistically significant effect of LTR on MMR. Despite the overall significance of the model (F-statistic = 676.1, p-value < 0.01), there are issues here as well with similar residual autocorrelation (Durbin-Watson = 0.532) that suggest that further refinement of the model may be needed to address potential model disparities.

Discussion/Conclusion This study analyzed global trends in maternal health from 2000 to 2023, focusing specifically on maternal mortality ratio (MMR), maternal deaths, and lifetime risk (LTR). While there has been significant improvement globally, regional disparities have remained. In most regions, MMR and maternal deaths have decreased, while LTR has generally increased, though with a slight decline toward the end.

When comparing regions like Africa and Europe/Central Asia, the differences are somewhat clear. African countries have high MMR and have seen little progress, while Europe and Central Asia have steadily reduced MMR. Conversely, maternal deaths in Africa were already low in 2000, so there hasn't been much change over the past two decades. East Asia and Middle East and North Africa show a clear improvement in maternal health, with declining MMR and maternal deaths. Conversely, West and Central Africa are an exception as we see maternal deaths increasing. These conversely related trends highlight the complex challenges some regions face in respect to maternal health whether that is due to limited skilled healthcare and regional wealth.

Although MMR and maternal deaths have improved, LTR has shown an upward trend across all regions, indicating that more women are at risk of dying from pregnancy-related causes over their lifetimes. The weak correlation between maternal deaths and LTR suggests that factors other than mortality may be influencing their risk, like previously discussed as healthcare access and economic conditions.

The multiple regression models for both Europe/Central Asia and Eastern/Southern Africa showed that maternal deaths strongly influence MMR, but no significant relationship was found between LTR and MMR. This suggests that while reducing maternal deaths helps lower MMR, the lifetime risk of maternal death is being influenced by other factors. The presence of residual autocorrelation in both models indicates that there may be unaccounted variables or trends affecting the results.

Overall, this analysis showed significant progress in reducing MMR and maternal deaths globally, but large disparities remain, particularly in African countries as expected from other analyses done globally. (Roser & Ritchie, n.d.). The increasing LTR across all regions suggests that there are still major challenges to address such as healthcare access and wealth that contribute to the regional and global inequalities. The regression analysis highlighted that maternal deaths play a key role in MMR, but LTR seems influenced by a broader set of factors.

This analysis highlights that more efforts are needed to improve maternal health outcomes, especially in low-resource regions. While some trends were assessed, one limitation in the analysis can be due to underreporting or gaps within the data, whether that is regional in specific countries or in time as biases in data collection can lead to inaccuracies. We also did not account for external influences such as political stability and or other social determinants such as health care and education which all affect maternal health. In further analyzes, it would be beneficial to consider the social and economic factors discussed to have a more rounded perception on all the complex factors improving or decreasing maternal health in all these regions. We could conduct studies where data for each death or risk is associated with data of individuals every day socio and economic situation to get a better holistic approach to understanding the global and regional trends of maternal health which establishes a better foundation for improvement measures for individuals.

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